

Gas Pressure, the Particle Model and Diffusion — Paper A

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
 - A calculator is allowed, but every number on this paper is designed to work without one.
 - Every numerical answer must carry a unit. An answer with no unit does not score the final mark.
 - Explanations must be written in terms of particles. “The particles get bigger”, “the particles expand” and “the particles push harder because they are squashed” score nothing.
 - The mark for each question is shown in square brackets on the right.
 - Reminder: pressure = force $\div$ area, measured in pascals (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$.
- Atmospheric pressure at sea level is about 100 000 Pa.

SECTION 1 – PRESSURE, FORCE AND AREA

- 1.** State what is meant by pressure, write down the equation linking pressure, force and area, and give the unit of pressure. [2]

- 2.** Calculate the pressure in each case.

- (a)** A force of 240 N acting evenly over an area of 0.4 m^2 . [1]

- (b)** A force of 60 N acting evenly over an area of 0.02 m^2 . [1]

- (c)** A force of 1500 N acting evenly over an area of 0.05 m^2 . [2]

- 3.** Calculate the force in each case.

(a) A pressure of 200 Pa acting on an area of 3 m². [1]

(b) A pressure of 100 000 Pa acting on an area of 0.02 m². [1]

(c) A pressure of 50 000 Pa acting on an area of 0.4 m². [1]

4. Describe, in terms of particles, what causes a gas to exert a pressure on the walls of its container. [2]

SECTION 2 – CHANGING THE PRESSURE

5. A sealed gas syringe is pushed in until the volume of the trapped gas is halved. The temperature does not change. State what happens to the pressure, and explain why in terms of particles. [3]

6. A sealed metal tin of gas is heated. The tin does not change shape, so the volume stays the same. State what happens to the pressure and give the two separate reasons for it, in terms of particles. [3]

- 7.** A gas is sealed in a cube-shaped box. Explain why the pressure on the lid, the base and all four sides is the same. [2]

- 8.** More air is pumped into a bicycle tyre, which does not get any bigger and does not get warmer. Explain why the pressure inside rises. [2]

SECTION 3 – SAYING IT CORRECTLY

- 9.** A student writes: “When you squash a gas the particles are pushed closer together, so they push harder on the walls.” State what is wrong with this and write a correct explanation. [3]

- 10.** A gas is heated from 27 °C to 327 °C inside a sealed container of fixed volume.

(a) Convert both temperatures into kelvin. [1]

(b) State, with a reason, what happens to the pressure. [1]

- 11.** State what is meant by diffusion. [2]

- 12.** State two changes that would make a gas diffuse faster, and explain one of them in terms of particles. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.